

Interpretation of the sfaEI_fast run (2026-09-14): where the paper's MATLAB results stand

Run: *data/sfaEI_fast* (analyses 53 min) and *figs/sfaEI_fast* (30 of 30 entries, 16 in-paper, 11 min; *report.md* and *manifest.md* there; transcript in *log.txt*). Preset *celltype_pairs_sfaEI_Sc0p2sig0p1_noise0p025_dualStd_3cond_mu8p*, the paper network with STD **unmatched** (TR's decision of 2026-09-14), run mode *fast*: 4 levels $\times$ 3 reps per 1-D sweep, 7 τ levels $\times$ 7 reps, 27 joint-sample points, 20-s simulations, 5 paired memory-capacity trials, 1 seed for the spectrum and transient-gain stages. Memory capacity ran with the Wiener process OFF on the SRA1 integrator (*cfg.mc_preset* chained to the paper preset with $\sigma_u = 0$, *cfg.mc_ode_solver* = 'sra1'). Where a medium-resolution number exists it is quoted from *data/topk_med* (same physics, 15 reps, 11 levels, 15 MC trials with noise ON), so the reader can see which fast numbers are already stable. Fast numbers are for direction and size; manuscript numbers come from the medium run of this bundle (*sfaEI_fast_config* with *run_mode* = 'medium').

1. The headline numbers

measure	none	single-timescale	multiple-timescale	source
λ_1 near the default point, median [IQR] (fast, 21 networks)	+3.65 [+0.07, +4.30]	+0.61 [-0.38, +1.40]	-0.118 [-0.136, +0.049]	fig_local_vs_finite_lle table
same at medium (105 networks)	+3.47 [+1.81, +4.33]	+0.35 [-0.06, +0.86]	-0.112 [-0.113, -0.110]	data/topk_med
networks with $\lambda_1 < 0$ at the default	24% / 4% (fast / medium)	29% / 31%	71% / 100%	same
share of time locally expanding (leading local rate > 0)	74% / 83%	54% / 53%	18% / 3%	same
Spearman $\rho(\lambda_1, \text{mean rate})$, 111 networks	-0.46 [-0.63, -0.25]	-0.47 [-0.63, -0.28]	-0.03 [-0.22, +0.14]	fig_lle_vs_rate
slowest τ_a sweep 1 $\rightarrow$ 30 s (E AND I moved), λ_1	-	-	-0.30 $\rightarrow$ -0.039, every rep negative	fig_sfa_EOC_allStd
top-K spectrum, 1 seed, noise on: n_+ / h_{KS} (bit/s) / D_{KY}	8 / 22.6 / 18.1	6 / 2.4 / 14.0	0 / 0 / 0	fig_lyapunov_spectrum
memory capacity, NOISE-FREE, SRA1, 5 trials: total MC	0.106 [0.096, 0.117]	0.48 [0.39, 0.57]	16.3 [15.3, 17.3]	fig_memory_capacity
memory horizon ($R^2 > 0.1$)	0 s	0.36 s	10.6 s [9.4, 12.1]	same
memory capacity, noise ON, 15 trials (medium)	0.103	0.259	0.590 (horizon 0.52 s)	data/topk_med
transient gain, active propagator, peak G / t_{peak}	1209 (still rising at 1 s)	30 (0.87 s)	8.6 at 0.18 s	fig_transient_gain
same, J_{xx} frozen (the conventional rate-network picture)	47 000	11 000	39	same
operating point: $\alpha(J_{xx}) / \alpha(J) / \lambda_1$	+8.7 / +8.7 / +4.0	+7.2 / +2.6 / +0.44	+1.8 / +1.1 / -0.11	same

measure	none	single-timescale	multiple-timescale	source
imbalance examples, λ_1 at $\mu_{EE} \times 0.5 / 1 / 1.5$	+1.98 / +3.94 / -4.23	-0.19 / +0.63 / -0.53	-0.10 / -0.11 / -0.11	fig_eig_heatmap_imbalance
mean rate at $\mu_{EE} \times$ 0.5 / 1 / 1.5	0.06 / 0.43 / 0.87	0.05 / 0.25 / 0.71	0.08 / 0.22 / 0.27	same

Numerics gate (fast, 2 seeds, 10-s windows): convergence checks A and B pass in every condition (order slopes 2.0, 2.0, 1.8); the Jacobian check J passes everywhere (relative error 10^{-10}); the finite-time LLE agreement checks L and C **pass for the multiple-timescale network** ($|\Delta\lambda_1|$ 0.006 between SRA1 and ode45; 0.005 Benettin vs QR) and **fail for the chaotic and intermittent regimes** (0.11 and 0.17 between integrators; 0.31 Benettin vs QR for single-timescale). That is not a defect of the integrator: over a 10-s window two integrators follow different trajectories of a chaotic system, and the finite-time exponent scatters by that much between trajectories. The gate is judged at medium (25 seeds, 20-s windows) and the Methods must state the claim at the level the data support (section 4).

2. What each result means

Stability (Results subsection 1). The multiple-timescale network sits at $\lambda_1 = -0.11$ in 100% of 105 medium networks and its top-K spectrum has no positive exponent ($h_{KS} = 0$, $D_{KY} = 0$). It is stable, not “slightly chaotic”. The stronger claim in Codex’s §3 (“transient expansion within long-interval stability”) is **not** what this regime does: its leading local rate is positive 3% of the time at medium (18% at fast, 1 seed per level). The regime that shows intermittency is the **single-timescale** one: locally expanding half the time, λ_1 straddling zero (31% of networks stable), which is exactly the edge-of-chaos picture. So the paper should say: multiple-timescale adaptation makes the network reliably and quietly stable; single-timescale adaptation leaves it intermittent at the boundary; no adaptation is chaotic ($\lambda_1 +3.5$, D_{KY} 18, h_{KS} 23 bit/s). Keep the “transient expansion” idea for the Discussion or attach it to the single-timescale regime.

Robustness across connectivity (subsection 2). The medians sheet (fig_sensitivity_medians, now with IQR bands and unclipped axes) shows the multiple-timescale curve flat at ≈ 0 across E:I ratio, network size, μ_{EE} , μ_{EI} and μ_{IE} , while the no-adaptation curve swings between -10 and +5 and the single-timescale curve between -4 and +1.5. The one axis that defeats it is μ_{II} : past +50% all three conditions go chaotic together (disinhibition). The imbalance heatmap sheet makes the same point in one picture: λ_1 pinned at -0.11 at half and 1.5 times the reference $E \rightarrow E$ mean, while the no-adaptation network goes from chaotic (+2, +3.9) to saturated-silent (-4.2). The “less sensitive” statement can now be a defined statistic: range of the median λ_1 over each sweep (multiple $\approx 0.3 \text{ s}^{-1}$ vs 5–14 for none), or the share of networks in a target band. The medium run of this bundle gives the manuscript’s values.

Rate and stability are not interchangeable (subsection 2). fig_lle_vs_rate: Spearman ρ between λ_1 and mean rate is -0.46 for no adaptation (an inverted U: quiet +0.6, mid +2.8, saturated -6.4), -0.47 for single, and **-0.03 for multiple** ($\lambda_1 \approx -0.11$ at every rate from 0.02 to 0.9). Adaptation decouples stability from operating point: quiet, mid-rate and saturated multiple-timescale networks are all equally stable. That is the sentence for the paper; do not use Pearson.

Long adaptation timescales (subsection 3). With BOTH cell types’ ladders swept (tau_levels.md proves it), λ_1 rises monotonically from -0.30 at $\tau_{slow} = 1 \text{ s}$ to -0.039 at 30 s, every rep negative, tracking $-1/\tau_{slow}$ (-0.033 at 30 s). The slowest SFA rung sets the exponent of the stable network. Note the change from the E-only sweep in data/topk_med (-0.112 at 1 s): moving the I ladder too makes the short- τ network more stable; the manuscript’s Methods already say E and I were swept, and now the data match.

Reservoir computation (subsection 4). Noise-free and on SRA1, the multiple-timescale network’s fading memory is **16.3** with a 10.6-s horizon, against 0.48 / 0.36 s for single-timescale and 0.11 / 0 for no adaptation; $d_z = -11$ for single vs multiple (p at the 5-trial floor of 0.0625; medium gives 15 trials and an exact test over 2^{15} patterns). The horizon of $\sim 10 \text{ s}$ is the 10-s SFA rung: the stable network holds the input in its slowest adaptation state. With the paper’s dendritic noise ON the same network gave MC 0.59 and a 0.52-s horizon, so the noise-free number is what the untuned recurrent dynamics can carry and the noisy number is what survives the network’s own fluctuations. Both should be reported; the noise-free one is the headline and the noisy one the caveat. Do not claim “requires little tuning” (nothing was tuned, but nothing measured how much tuning would have been needed).

Non-normal transient amplification (new, for subsection 1 or the Discussion). A unit perturbation of the dendritic states is amplified $8.6\times$ within 0.18 s in the stable multiple-timescale network before decaying (back to 1 by ~ 2 s at the 3-s horizon of the earlier note). The conventional frozen- J_{xx} picture predicts $39\times$ and, in the adapted regimes, a runaway ($\alpha(J_{xx}) = +1.8$ at a typical state); adaptation’s feedback at the operating point brings the frozen rate to $+1.1$ and the trajectory’s own rate is -0.11 . The amplifying pattern is a ~ 70 -neuron E/I-difference pattern (balanced amplification), unaligned with the Lyapunov direction ($|\cos| \approx 0.05$); the E/I sum mode is damped, and the isotropic (noise-average) gain is ≈ 1 . Reading for the paper: adaptation is part of the recurrent dynamics, not a correction to them; instantaneous Jacobian spectra (the occupancy heatmaps) describe the local expansion that the trajectory then removes. Full account: docs/notes/Non_normal_amplification.md, Transient_gain_frozen_vs_active.md, Transient_gain_summary_2026-09-13.md.

Jacobian occupancy (subsection 1 support). The heatmaps are now a defined comparison (three conditions, shared seeds, three E:I balances, matched λ_1 and mean rate in every panel). The median spectral abscissa of J_{xx} at sampled states is $+1.8 \text{ s}^{-1}$ in the stable regime: the instantaneous spectrum crosses zero while λ_1 is -0.11 , which is the caption’s point that occupancy is a local description and λ_1 the stability test.

3. The STD strength question, and what we can still say

Codex’s §1 was right that the one- vs multiple-timescale comparison changes depression STRENGTH along with timescale count (the two depression variables at the same $\tau_{\text{rel}}/\tau_{\text{rec}}$ ratio square the steady-state depression). Overnight three matchings were built and run at fast (docs/notes/STD_strength_matching_2026-09-13.md §5):

matching	multiple-timescale λ_1 near default	MC (noisy, 5 trials)
unmatched (this run, the paper)	-0.118	0.59 (medium, 15 trials)
dual routes scaled $\times 3$ (equal output at $r = 0.25$)	$+2.24$, chaotic	0.10
dual usage weakened ($\rho_u = 0.34$)	$+1.01$, chaotic	0.15
single route strengthened (τ_{rel} 0.25 $\rightarrow$ 0.0625), dual as published	-0.118 (unchanged)	0.58; single rises 0.24 $\rightarrow$ 0.39

TR’s decision: the manuscript keeps the unmatched comparison. What can honestly be said:

1. The comparison is between three **adaptive architectures** as they would be built (no adaptation; one SFA + one STD timescale; three SFA + two STD timescales), not between timescale counts at fixed strength. The Methods sentence that depression is deliberately not normalised stays, with one added clause: the two-timescale condition therefore also has stronger steady-state depression, and the comparison is of the combined timescale-plus-strength architecture. Do not write that the difference is caused by timescale count alone.
2. The matched runs are a supplementary control that supports a bounded version of the timescale claim: with the single-timescale depression strengthened to the dual’s steady state, the single-timescale network gains fading memory (0.24 $\rightarrow$ 0.39) but the multiple-timescale network still holds more (0.58) and stays stable; and weakening the two-timescale depression to the single’s strength makes the network chaotic. So both strength and temporal structure contribute. That sentence can go in the Results or Discussion with the matched runs as supplementary figures once they are at medium.
3. The fig_STD_steady_state panel already in the supplement shows the squared depression; it should be captioned as such rather than as a matched pair.

4. Against the manuscript, section by section

Results 1 (stability boundary). Ready: fig_example_timeseries (composite with local rate and accumulating λ_1), fig_eig_heatmap / fig_eig_heatmap_imbalance, fig_lyapunov_spectrum, fig_local_vs_finite_lle. Numbers to quote from the medium run. Text change: drop the transient-expansion framing for the multiple-timescale regime; attach it to the single-timescale one; add the non-normal transient paragraph.

Results 2 (connectivity robustness). Ready: the sensitivity sheets with bands, fig_EI_weights_param_space, fig_lle_vs_rate (new), fig_eig_heatmap_imbalance (new). Text change: state “less sensitive” as a range or a share, quote Spearman not Pearson, and add the μ_{II} exception.

Results 3 (long timescales). Ready: `fig_sfa_EOC_allStd` now unclipped, on both cell types, with `tau_levels.md`. The image link in the manuscript still points at the old folder name; the tag is unchanged.

Results 4 (recurrent computation). Ready after the medium run: `fig_memory_capacity` redrawn from saved statistics, with provenance. Text change: the noise-free protocol must be stated ($\sigma_u = 0$, SRA1, otherwise identical), the noisy result kept as a caveat, the 10-s horizon tied to the 10-s SFA rung, and the “prospective” PyTorch sentence left as is (that repository was not touched).

Results 5 (stimulation). Nothing changed. `fig_stim_engages_adaptation` still runs on the separate hand-tuned bursting preset with one seed; Codex §8–§9 remain open.

Methods / Supplemental Methods. Now backed by saved artifacts: SRA1 vs ode45 step refinement (A), strong-order convergence on one Brownian path (B), the analytic Jacobian vs central finite differences (J), Benettin vs QR vs top-K (C). The LLE-agreement claims (L, C) hold in the stable regime and must be qualified for the chaotic regimes: agreement to $\sim 0.1\text{--}0.3\text{ s}^{-1}$ on 10-s windows, i.e. within the finite-time scatter between trajectories; the medium run is the test. The Methods’ statement that the Lyapunov exponent is Benettin’s needs updating: the sweeps use the top-K QR method ($K = 15$, retried to 60) and Benettin is the cross-check. The τ sweep description (E and I) is now true.

5. Against `fig_to_do_in_future.md` (Codex’s ranked list)

§	item	status
0	verbose setting	done (three levels, minimal default)
1	freeze the 1-TS vs MTS comparison, STD normalisation	decided: unmatched stays; three matchings built, run at fast, documented; one sentence of scope in the Methods, matched runs as a supplementary control
2	numerical-validity gate	built (five checks, pre-registered thresholds, verdict file, Jacobian artifact); passes for the stable regime at fast; judged at medium
3	main stability result with local + finite-time LLE across trials	figures built; the across-trial evidence says the stronger claim belongs to the single-timescale regime
4	robustness + rate vs stability	figures built (bands, <code>fig_lle_vs_rate</code>); needs the medium rerun for the 15-network numbers
5	longest-timescale sweep on both E and I, unclipped	done; medium rerun pending
6	memory capacity audit and redo	figure and provenance done; noise-free protocol adopted; 15-trial medium run pending (30-trial production optional)
7	Jacobian occupancy as a defined comparison	done (<code>fig_eig_heatmap_imbalance</code>); medium at $n = 500$ pending
8–9	bursting / stimulation redesign, DC-LLE in the bursting net	not started
10	the two composite Methods/Intro figures	the time-series half of figure 2 is done (<code>fig_example_timeseries</code>); the Sompolinsky intro figure and the assembly are not
11	pulse trains	not started, by design

§	item	status
12	final assembly and consistency checks	after the medium run

6. Against the two reports

reality_manuscript_suggested.md status changes: SRA1-vs-ode45 and step-halving are now “reproducibly verified” with an acceptance record; the Jacobian finite-difference check is automated and saved; Benettin vs QR exists as check C with the qualification above; the longest-timescale sweep on both cell types is real; the memory-capacity audit is done and the protocol changed to noise-free. Still prospective: PyTorch three-condition run, the CSCS analyses, the absolute figure path, per-figure provenance (partly: each figure root now has manifest.md, command_window.log and the tables).

suggest_updates_main_matlab_model.md: its recommended option 1 (keep the product model, add a strength-matched control) is what the matched presets are; the result is that the control changes the conclusion, which is why the paper’s claim must be stated at the architecture level. Its Jacobian-occupancy recommendations are implemented. The bursting and PyTorch recommendations are untouched.

7. Message to the manuscript agent

1. Keep the three conditions and the unmatched comparison. State in the Methods that the multiple-timescale condition has both more timescales and stronger steady-state depression, and phrase every contrast as between adaptive architectures. Use the matched runs (when at medium) as a supplementary control with the two-sentence summary in section 3 above.
2. Results 1: the multiple-timescale network is stable ($\lambda_1 = -0.11$, no positive exponents, locally expanding 3% of the time); intermittent transient expansion is the single-timescale regime’s property. Add the non-normal transient amplification paragraph ($8.6\times$ at 0.18 s; the frozen picture overstates it and predicts a runaway).
3. Results 2: quote the range of the median λ_1 per sweep, the μ_{II} exception, and Spearman ρ of $-0.46 / -0.47 / -0.03$; add fig_lle_vs_rate and fig_eig_heatmap_imbalance.
4. Results 3: unchanged claim, now on both cell types, with the $-1/\tau$ reading.
5. Results 4: noise-free SRA1 protocol; headline MC and horizon from the medium run; noisy result as the caveat; horizon = the slowest SFA timescale.
6. Methods: top-K QR is the estimator, Benettin the check; the numerics claims are backed by numerics_verdict.md, with the finite-time qualification for the chaotic regimes; the τ sweep moves E and I.
7. Every number in the text comes from the medium run of sfaEI_fast_config (with run_mode = 'medium'), not from this fast run.

8. What is pinned down, what changed, what is next

Pinned down: the sign and size of λ_1 in the three regimes; the flatness of the multiple-timescale network across five of six connectivity axes and across E:I balance; the $-1/\tau$ dependence on the slowest SFA rung; the decoupling of rate and stability under multiple-timescale adaptation; the order of the memory-capacity result (three orders of magnitude noise-free); the transient-gain picture; the numerical convergence and the Jacobian.

Changed since the ranked list was written: the STD-matching question is answered rather than open (and the answer is to keep the unmatched comparison with honest scope); memory capacity is measured noise-free on SRA1; the τ sweep moves both cell types; the sweeps store local rates and the transient-divergence scalars; every entry point logs its transcript; the model has a per-route weight scale (unused by the paper preset).

Next: the medium run of this bundle (about two hours plus memory capacity), then the numerics verdict at 25 seeds, then the figure assembly. The matched presets at medium are optional supplementary material. The bursting/stimulation redesign (§8–9) is the one Results subsection with no new evidence.